Phishing and Ransomware Scenario Simulation Lab

STUDENT WORKBOOK

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1. Overview

This workbook and runbook provide a combined learning and operational guide for conducting a phishing-based ransomware simulation in a controlled virtual laboratory environment. The document is designed to guide students through both the conceptual understanding and practical execution of the lab while also serving as a technical reference for repeatability. The simulation demonstrates how phishing can be used as an initial attack vector and how such an attack may escalate into ransomware-like behavior, using a benign and ethical approach.

2. Learning Objectives

By completing this lab, students will be able to:

- Understand phishing as an initial attack vector
- Deploy a basic phishing campaign using GoPhish
- Simulate ransomware behavior safely using Powershell
- Observe host-based and network-based logs
- Document findings in a structured incident report
- Reconstruct an incident timeline from collected evidence

3. Lab Environment

3.1 Virtual Machines

VM Name	Operating System	Role
Victim	Windows 10 Pro	End user workstation
Utility	Debian 12	Mail server, IDS, logging
Attacker	Kali linux	Phishing platform

3.2 Network Configuration

- Host-only networking for internal communication
- NAT enabled where required for package installation

4. Tools

- ✓ **Gophish** – Phishing Framework
- ✓ **Postfix and Dovecot** – Email delivery and access
- ✓ **Suricata** – Intrusion Detection
- ✓ **Powershell** – Ransomware simulation
- ✓ **Wireshark and tcpdump** – Network traffic analysis
- ✓ **Windows Event Viewer** – Host-based logging

5. Lab Instructions/Tasks

Task 1: Environment Preparation

1. Power on all virtual machines.
2. Verify network connectivity between VMs.
3. Take snapshots of all machines (`Pre-Attack`).
4. Ensure required services are installed.

Task 2: Start Required Services

Debian Utility VM

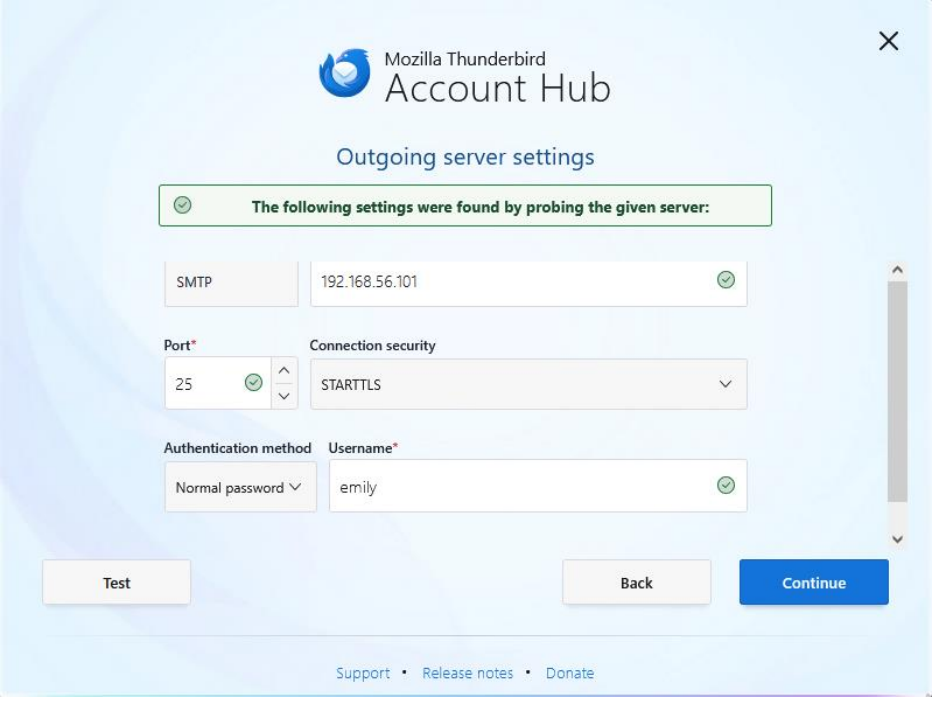
- sudo systemctl start postfix
- sudo systemctl start dovecot
- sudo systemctl start suricata

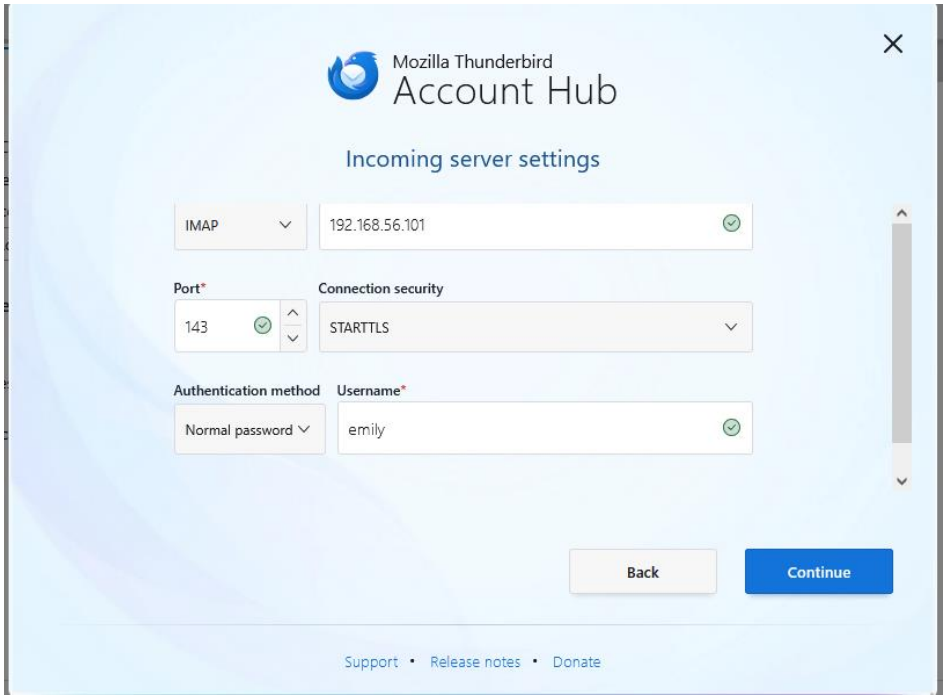
Verify if services are running

sudo systemctl status postfix dovecot suricata

Task 3: Set up user Thunderbird's account

1. Configure Thunderbird Outgoing/Incoming Server Settings





Task 2: Phishing Email Setup

1. Update and Upgrade Kali Linux

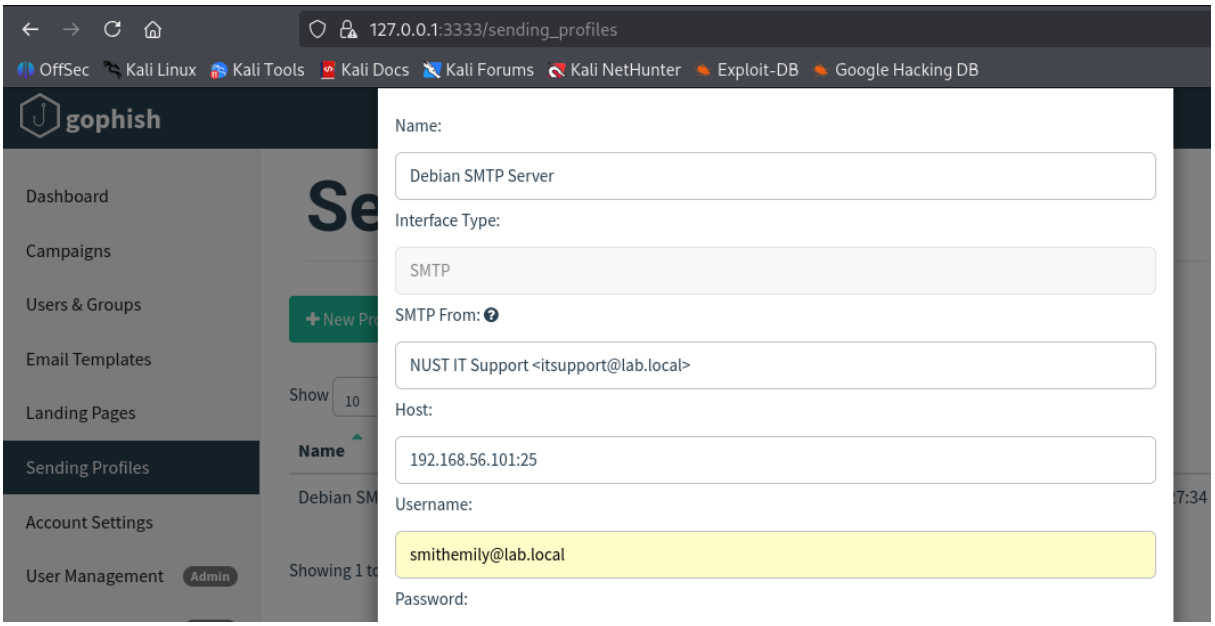
```
┌──(uendjipa㉿attacker)-[~]
└─$ sudo apt update && sudo apt upgrade -y
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.6 MB]
45% [3 Contents-amd64 6,923 kB/52.6 MB 13%]                    3,919 kB/s 12s
```

1.1 Install and Launch GoPhish on Kali Linux.

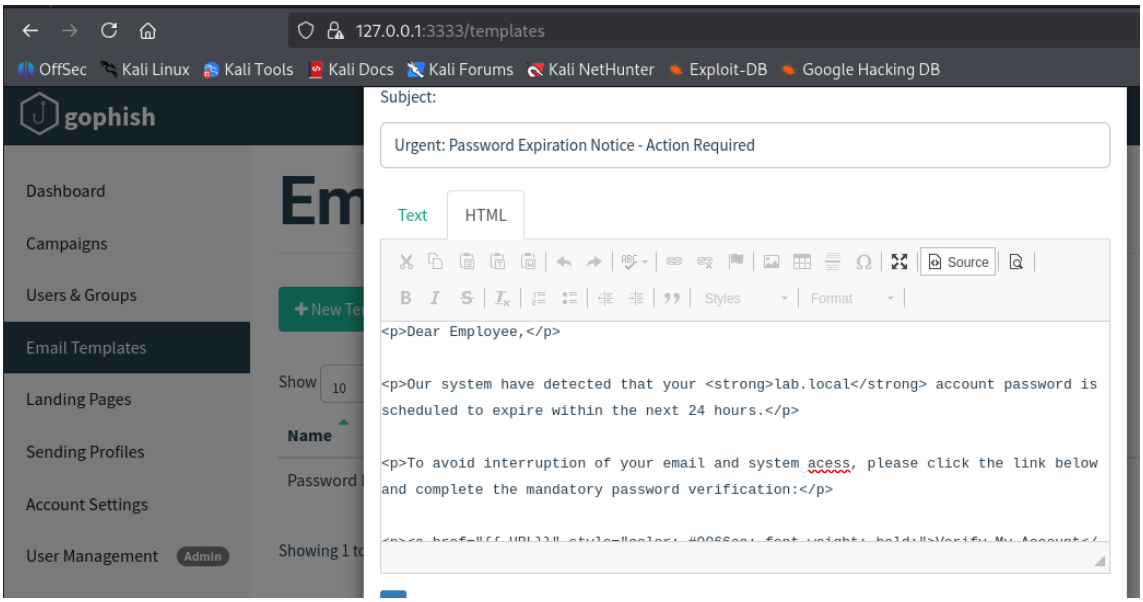
```
uendjipa@attacker: ~

┌──(uendjipa㉿attacker)-[~]
└─$ sudo apt install gophish
[sudo] password for uendjipa:
gophish is already the newest version (0.12.1-0kali3+b1).
gophish set to manually installed.
The following packages were automatically installed and are no longer required:
  amass-common                libtag1v5
  firebird3.0-common          libtag1v5-vanilla
  firebird3.0-common-doc      libtagc0
  firmware-ti-connectivity    libtheora0
  icu-devtools                libunwind-19
  libabsl20230802             libutempter0
  libbfio1                    libvpx9
  libbson-1.0-0t64            libwireshark18
  libc++1-19                  libwiretap15
  libc++abi1-19               libwsutil16
  libcapstone4                libx264-164
  libconfig++9v5              libyelp0
  libconfig9                  openjdk-23-jre
  libegl-dev                  openjdk-23-jre-headless
  libflac12t64                python3-appdirs
  libfmt9                     python3-bluepy
  libgail-common              python3-click-plugins
  libgail18t64                python3-dunamai
```

2. Create a sending profile using the internal mail server.

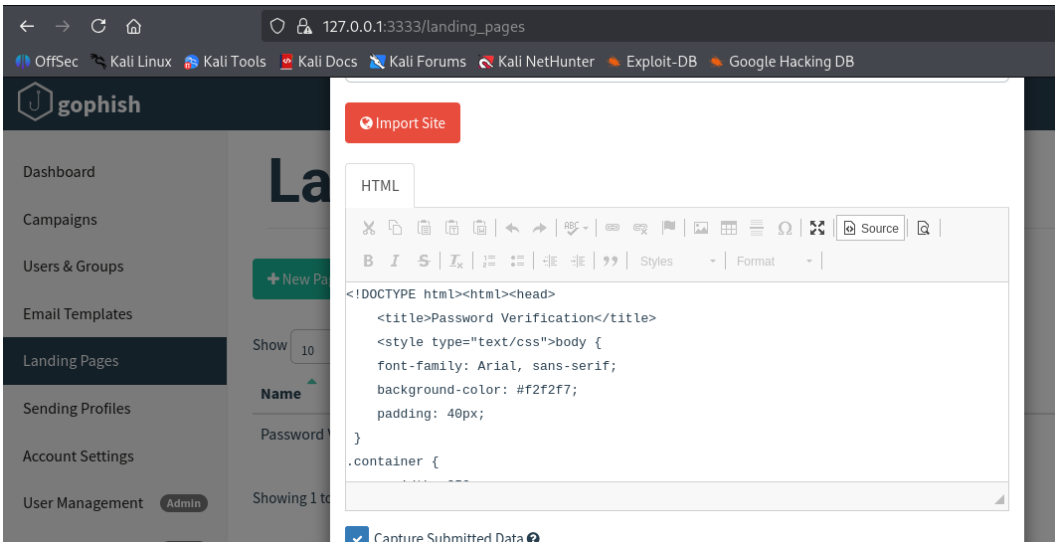


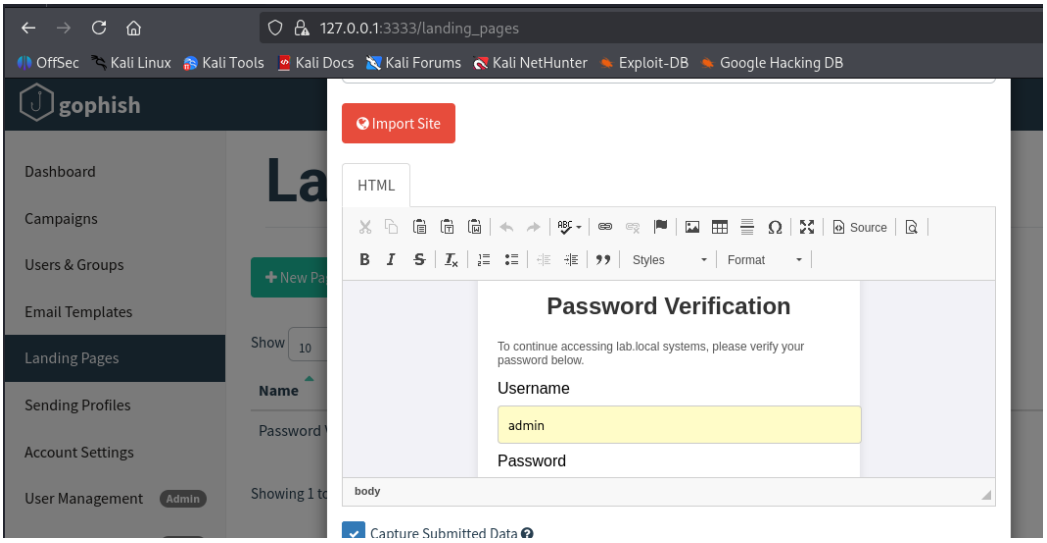
3. Design a phishing email template.



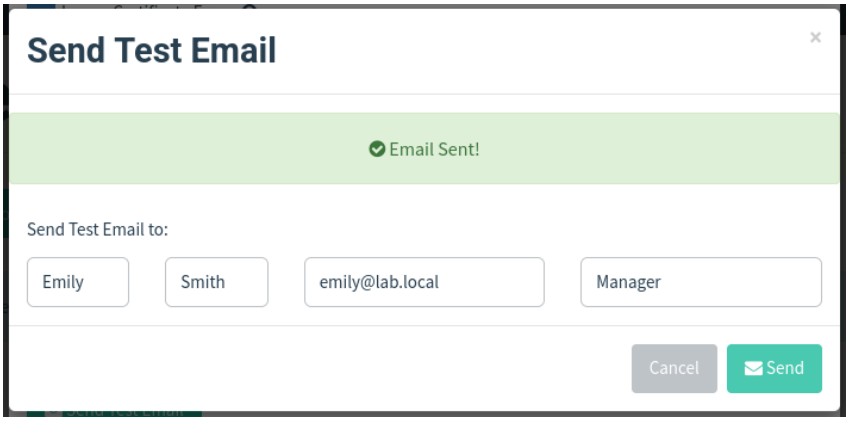
Tip: Add a few grammar and spelling mistakes to give the phishing email a realistic effect. Real world phishing emails tend to have some grammatical mistakes.

4. Create a landing page to simulate credential capture.



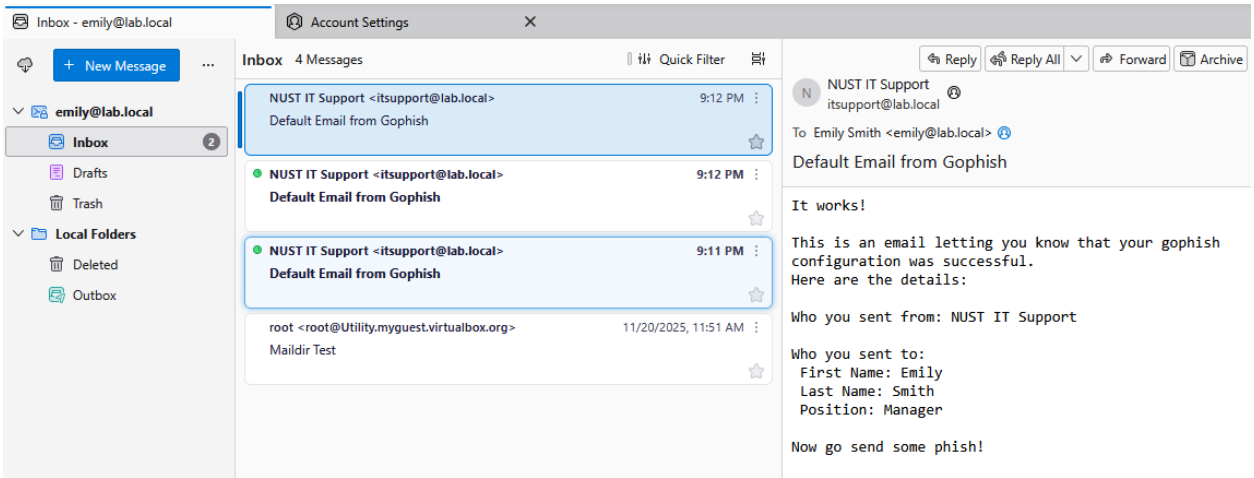


5. Verify email delivery to the test user.

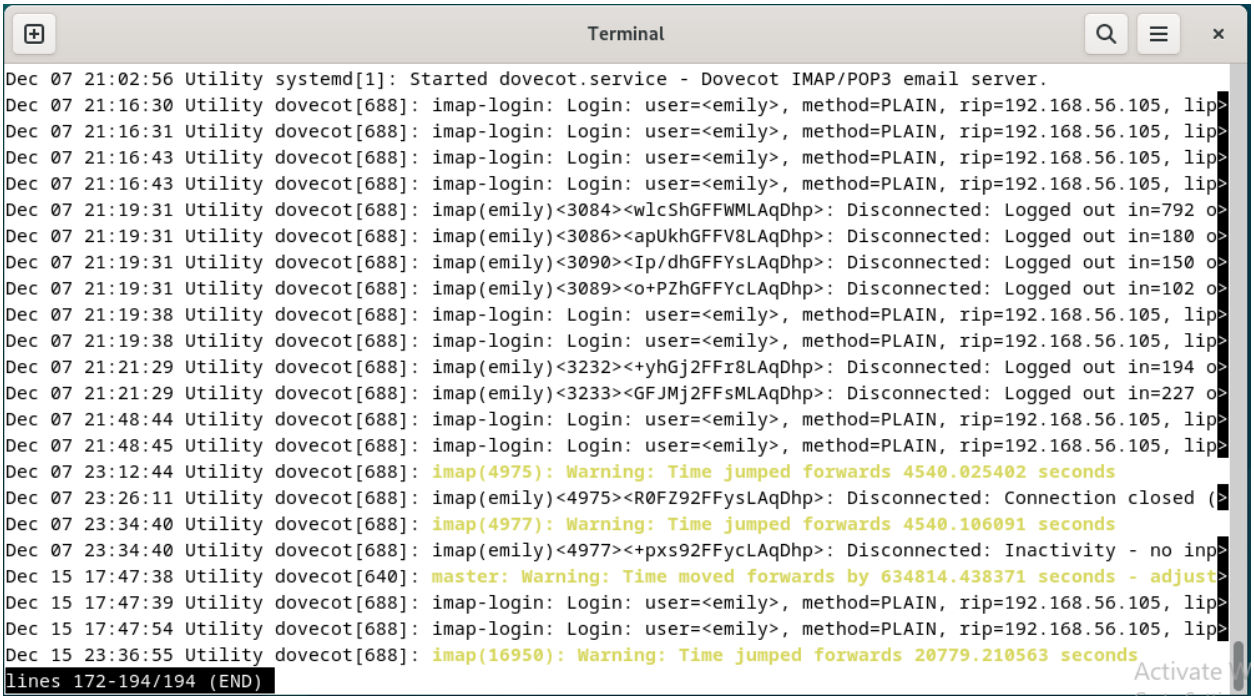


Task 3: Email Delivery Verification

1. Log into the test user mailbox using Thunderbird.
2. Confirm receipt of the phishing email.



3. Document mail server activity on Debian.

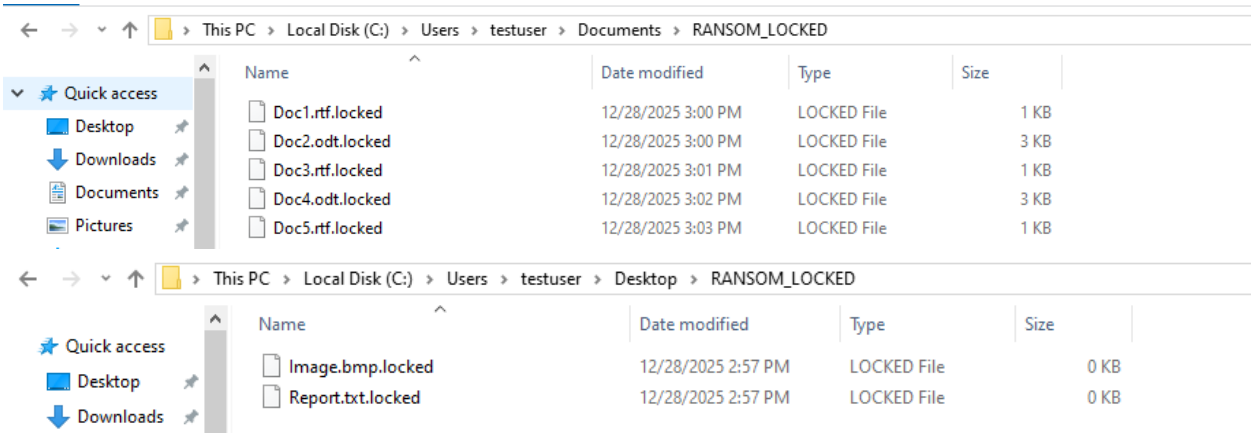


Task 4: Benign Ransomware Simulation

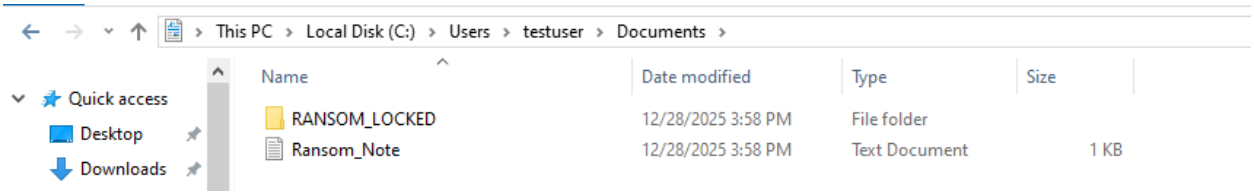
1. Log into the Windows victim machine.
2. Execute the provided PowerShell script.

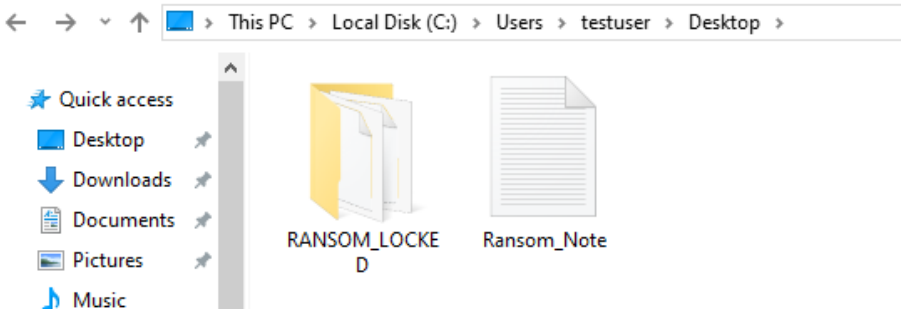
```
PS C:\Users\testuser> powershell -ExecutionPolicy Bypass -File C:\Tools\simulate_ransom.ps1
Starting benign ransomware simulation...
[+] Locked: doc1.txt from C:\Users\testuser\Documents
[+] Locked: doc2.txt from C:\Users\testuser\Documents
[+] Locked: doc3.txt from C:\Users\testuser\Documents
[+] Locked: doc4.txt from C:\Users\testuser\Documents
[+] Locked: doc5.txt from C:\Users\testuser\Documents
[+] Locked: Image.bmp from C:\Users\testuser\Desktop
[+] Locked: Report.txt from C:\Users\testuser\Desktop
Simulation complete. Ransom notes created in Documents and Desktop.
PS C:\Users\testuser>
```

3. Observe files being locked in the Desktop and Documents directories.



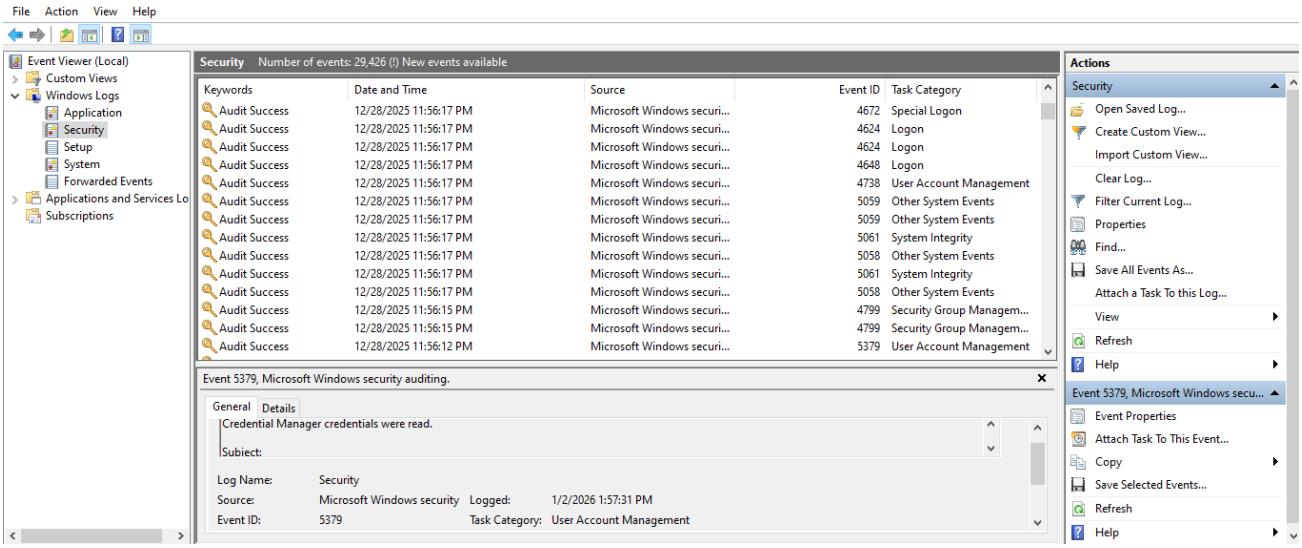
4. Verify ransom note creation

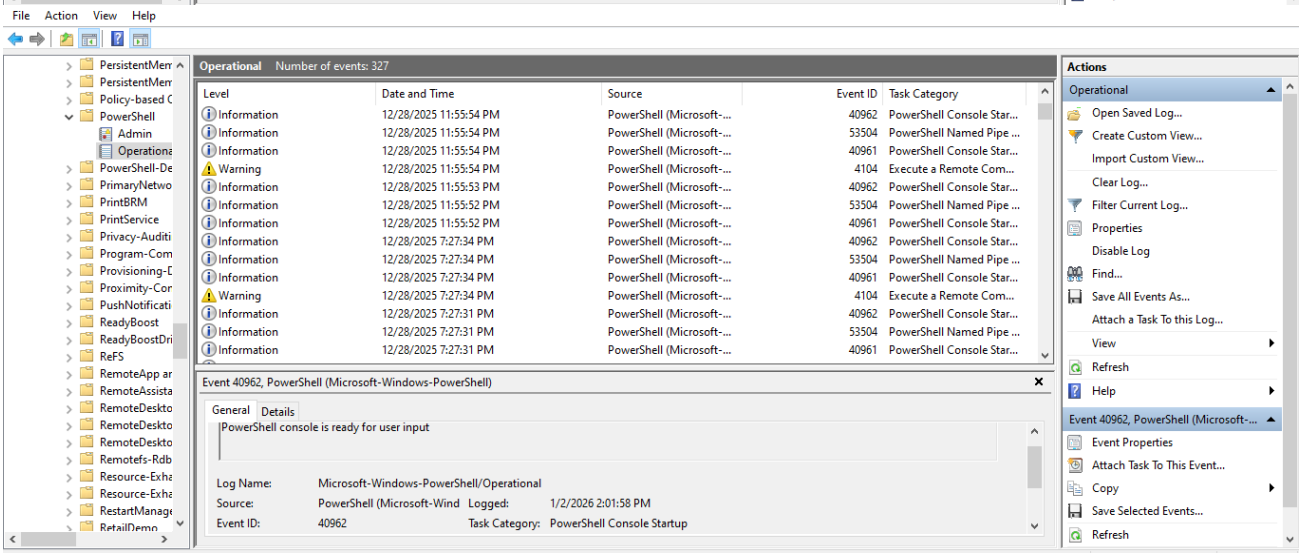




Task *5*: Detection and logging

1. Review Windows Security and PowerShell (Operational) logs.





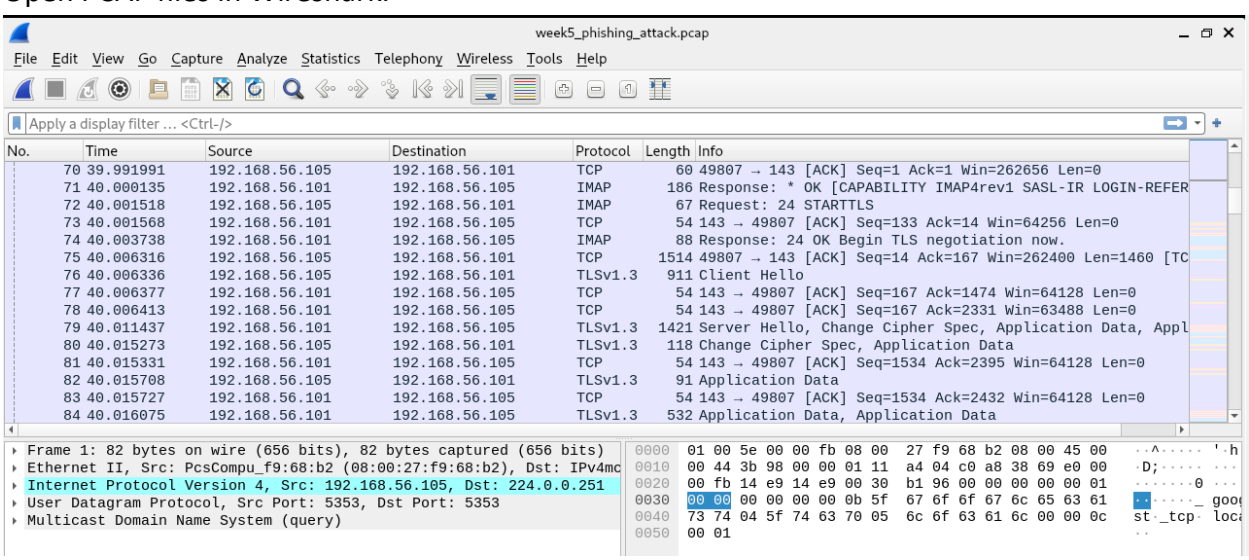
2. Inspect Suricata logs on Debian.

```
Terminal
root@Utility:/home/emily# sudo nano /etc/suricata/rules/custom.rules
root@Utility:/home/emily# sudo tail -f /var/log/suricata/fast.log
12/18/2025-22:05:44.454171  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:09f8:2061:af6f:1341:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
12/18/2025-22:07:29.229868  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:10:44.537467  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:09f8:2061:af6f:1341:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
12/18/2025-22:11:54.719000  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:12:55.722501  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:13:56.715271  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:14:57.725675  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:15:44.667116  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:09f8:2061:af6f:1341:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
12/18/2025-22:15:58.741430  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:16:59.767266  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
12/18/2025-22:19:05.264923  [**] [1:9000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3]
{IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a720:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
```

3. Capture network traffic using tcpdump.

```
Terminal
emily@Utility:~$ su
Password:
root@Utility:/home/emily# tcpdump -r week5_phishing_attack.pcap host 192.168.56.105
reading from file week5_phishing_attack.pcap, link-type EN10MB (Ethernet), snapshot length 262144
19:27:03.717287 ARP, Request who-has Utility tell 192.168.56.105, length 46
```

4. Open PCAP files in Wireshark.



Task 6: Incident Analysis

1. Correlate timestamps across logs.

- Gophish logs
- Postfix logs
- Dovecot logs
- Windows Security logs
- Windows Powershell logs

2. Identify the attack progression.
3. Reconstruct a simple incident timeline.

Time	Event	Source
14:35:12	Phishing email sent	Gophish
14:35:40	Email delivered	Postfix logs
19:27:03	Email accessed	Dovecot logs
23:55:52	Payload Executed	Powershell output
23:55:54	Files locked	Windows file system
23:55:05	Network traffic observed	PCAP

6. Expected Outcomes

- Successful delivery of phishing email
- Execution of simulated ransomware behavior
- Generation of host and network logs
- Ability to explain the attack flow and impact

7. Safety and Ethics

This lab uses a benign simulation script and is conducted in an isolated environment. No real malware is deployed, and no external systems are affected.

8. Optional Exercises

- Tune Suricata rules to detect phishing-related traffic
- Create PowerShell script to simulate recovery
- Add centralized logging using ELK

9. Conclusion

This lab successfully demonstrated how a phishing attack can act as an entry point for a ransomware-style incident within a controlled virtual environment. By completing the practical tasks, students were able to observe the full attack lifecycle, from phishing email delivery and user interaction to simulated payload execution and evidence collection. The use of a benign PowerShell script ensured that the exercise remained ethical while still providing realistic insight into ransomware behavior.

Through hands-on interaction with tools such as GoPhish, Postfix, Dovecot, Suricata, and Wireshark, students gained practical experience in both offensive and defensive cybersecurity concepts. The lab emphasized the importance of monitoring, logging, and incident analysis in detecting and responding to security incidents. Additionally, students learned how environmental constraints, such as limited system resources, can influence the design and implementation of security solutions. The skills developed through this exercise prepare students to better understand, detect, and respond to phishing-based attacks in professional settings.